

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Mechanical Engineering Semester : III Subject Code & Name: Engineering Mathematics – III (BTBSC301) Max Marks: 60 Date: / /2022 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Find $L\{f(t)\}$, where $f(t) = t^2 e^t \cos 2t$	Understand	6
B)	Find Laplace transform of periodic function $f(t) = E \sin \alpha t$ $0 < t < \frac{\pi}{\alpha}$ and $f(t) = f(t + \frac{\pi}{\alpha})$	Understand	6
C)	Using Laplace transform evaluate $\int_0^\infty e^{-2t} \frac{\sin^2 t}{t} dt$	Evaluation	6
Q.2	Solve Any Three of the following.		12
A)	Find $L^{-1}\left\{\log\left(\frac{s+a}{s+b}\right)\right\} = \dots\dots$	Application	4
B)	Find $L^{-1}\left\{\frac{s^2}{(s^2+3^2)(s^2+2^2)}\right\}$ using Convolution theorem.	Application	4
C)	Using Laplace transform solve $y'' + y = 6 \cos 3t$, $y(0) = 3$, $y'(0) = 1$	Application	4
D)	Solve $\frac{dx}{dt} = 2x - 3y$ and $\frac{dy}{dt} = y - 2x$, given $x(0) = 8$, $y(0) = 3$	Application	4
Q. 3	Solve Any Two of the following.		12
A)	Find the Fourier integral representation of the function : $f(x) = 1, \quad x \leq 1$ $= 0, \quad x > 1$ Hence evaluate 1. $\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$, 2. $\int_0^\infty \frac{\sin \lambda}{\lambda} d\lambda$	Evaluation	6
B)	Find the Fourier Cosine integral representation of the function : $f(x) = x^2, \quad 0 < x < a$ $= 0, \quad x > a$	Understand	6
C)	Using the Fourier Sine integral representation show that $e^{-2x} - e^{-3x} = \frac{10}{\pi} \int_0^\infty \frac{\lambda \sin \lambda x}{(\lambda^2+3^2)(\lambda^2+2^2)} d\lambda$	Understand	6
Q.4	Solve Any Three of the following.		12
A)	Form the partial differential equation by eliminating arbitrary function	Understand	4

	<i>f from $f(xy + z^2, x + y + z) = 0$</i>		
B)	Solve the partial differential equation $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$	Analysis	4
C)	Solve $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$ subject to the condition $u(0, t) = 0 = (\pi, t)$ $u(x, 0) = 2 \sin 3x - 4 \sin 5x$	Application	4
D)	Solve $\frac{\partial^2 z}{\partial x^2} = 4 \frac{\partial^2 z}{\partial y^2}$ satisfying the condition $z = \frac{\partial z}{\partial y} = \sin x$, when $y = 0$	Application	4
Q. 5	Solve Any One of the following.		12
1.A	If $f(z)$ is analytic function and $f'(z) = 0$ then prove that $f(z)$ is a constant function.	Understand	6
1.B	Evaluate $\oint \frac{z+4}{z^2+2z+5} dz$ Over C, Where C is the circle $z + 1 - i = 2$	Evaluation	6
	OR		
2.A	If $\phi = -x^3 + 3xy^2$ is flux function is an electrostatic field, find potential function and hence complex potential function.	Understand	6
2.B	Find the Residues of $f(z) = \frac{\sin z}{z \cos z}$ at its poles inside the circle $z = 2$	Understand	6
	*** End ***		

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